

The Arithmetic of Time: How Pensions Quietly Reshaped Generational Power

The 2025 Autumn Governance Series: Pension Fund Governance Focus



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OCT 29, 2025

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The Math Behind Trust

Discount rates are the quiet architects of modern power. They decide how pensions are valued but how entire generations inherit the costs. For decades, these actuarial levers have shifted trillions of dollars between workers and retirees, taxpayers and governments, the young and the old.

The math looks benign: a pension plan valued at a 7% discount rate becomes roughly 50% more expensive when re-priced at 4%. Yet behind that arithmetic lies one of the largest peacetime wealth transfers in history. A single percentage point embeds a philosophy of fairness—whether or not the market sees it.

The Arithmetic of Time

Pension systems translate promises into present value. They take what retirees will receive and divide it by a discount rate that represents how the present values the future. A higher rate shrinks tomorrow's obligation; a lower rate inflates them.

Callout: The Arithmetic of Time

A 15-year pension valued at a 7% discount rate becomes **45% more expensive at 4%.**

One line of math can move billions.

When governments and corporate plans assume unrealistically high real rates, they make benefits appear affordable even when the underlying economics disagree. The Society of Actuaries warned as early as 2013 that this effort subsidizes current generations at the expense of the next. That silent subsidy flows month after month, unnoticed but immense.

Between 2008 and 2022, falling interest rates increased global pension liabilities by an estimated 40–60% (OECD, *Pension Markets in Focus*, 2022). The numbers changed because time itself was re-priced. Monetary policy became generational policy.

When Central Banks Became Pension Architects

After the 2008 crisis, central banks cut rates to stabilize economies and launched quantitative easing. Yields collapsed. The U.S. 10-year Treasury fell from nearly 5% to below 1%. That single digit quietly rewrote the world's balance sheets.

Corporate pension funds saw liabilities soar even as investment portfolios stagnated. The funded ratio of America's largest corporate plans plunged from 106% in 2007 to 64% by 2009 (Milliman Corporate Pension Funding Index). A retiree promised \$40,000 annually didn't lose a cent, but the present value of that promise nearly doubled. Future workers absorbed the cost through lower wages, higher contributions, or smaller benefits.

In the U.K., low rates encouraged "liability-driven investment" strategies. When yields spiked in 2022, those funds faced £100 billion in margin calls. The Bank of England intervened to prevent systemic collapse—a crisis born of good intentions and bad duration math.

Across Europe, compressed discount rates triggered benefit cuts and funding shortfalls. The Netherlands' Ultimate Forward Rate fell from 4.2% to 2%, erasing solvency cushions. Japan's long experiment with zero interest rates pushed its public fund into equity and currency risk, shifting volatility to younger savers.

Everywhere, the pattern rhymed: older generations were protected by younger ones by hope.

Governance as Moral Arithmetic

Boards often treat discount rates as technical details. They are anything but. A discount rate is a moral claim disguised as a number—it declares how the future matters.

When trustees base assumptions on expected returns, they embed optimism into solvency. When they use risk-free rates, they anchor their promises in realism. The gap between those choices defines the honesty of a pension system.

Callout: The Governance Lens

Three questions every board should ask:

- Who bears the downside if these numbers are wrong?
- How long will it take to correct course once the error appears?
- Which generation is carrying the unspoken risk?

Canada's 1997 pension reform shows what that looks like. By pre-funding and raising contributions early, and insulating investment decisions from politics, the Canada Pension Plan achieved steady-state sustainability even under conservative assumptions. Transparency—not luck—built that durability.

In contrast, Illinois' assumption tweaks conjured billions in phantom savings. In Southern Europe, decades of under-funding turned pension systems into fiscal minefields. Governance faltered not through corruption but through

temporal blindness—the inability to confront long-term arithmetic in time.

Generational Accounting and the Hidden Contract

For early- and mid-career professionals, pensions can seem distant—a bureaucratic line on a pay stub. But every deduction represents an implicit social contract: you finance today's retirees with the expectation that someone will do the same for you.

That contract only endures if the math is honest. Social Security's interest rate of return has fallen from nearly 5% for those born in 1920 to about 1% for later cohorts (U.S. SSA data). Younger generations will pay more and receive less—not by malice, but by demographic logic.

Early participants in pay-as-you-go systems earned double the value of their contributions. Later generations break even at best; some will face negative real returns as population growth slows. The past received a premium; the future inherits the debt.

In corporate plans, aggressive discount assumptions inflate earnings today and shift liabilities tomorrow. When those projections miss, shareholders and younger workers both absorb the loss. Governance, in that sense, isn't just about compliance—it's about intertemporal fairness.

Lessons from Systems That Endured

Some countries have engineered stability into their systems. Sweden's notional defined-contribution model adjusts benefits automatically with demographic changes.

growth and life expectancy, letting the math self-correct before crisis. Netherlands' new collective DC scheme allocates investment returns by age group, explicitly balancing upside and downside risk. Canada's CPP Investment Board, prohibited from buying government debt, embodies structural independence as fiduciary safeguard.

Comparative Governance Snapshot

Automatic stabilizers are the quiet architects of fiscal resilience. They are the pre-programmed mechanisms that automatically adjust government spending and taxation in response to economic conditions, without the need for new legislation. These systems are designed to cushion the impact of economic downturns, ensuring that the government's budget remains balanced even in the face of uncertainty. They are the unsung heroes of fiscal policy, working behind the scenes to maintain the stability of the financial system.

These systems prove a simple rule: automatic stabilizers outperform political promises. Those that corrected early now adjust gradually; those that waited will face abrupt reckoning.

The Corporate Mirror

Corporate boards mirror these dilemmas. The discount rate they approve determines not only financial statements but the company's moral balance sheet.

When firms close defined-benefit plans and shift to defined-contribution schemes, they transfer longevity and market risk to individuals. Few employees recognize this shift in time preference; fewer still can hedge. Transparency about the real cost of promises isn't bureaucracy—it's

reputational capital.

Boards that govern pension risk effectively:

- Disclose liabilities under multiple scenarios—actuarial, market, and r—so stakeholders can see the range of futures.
- Link executive incentives to long-term funding health rather than an returns.
- Treat data and model governance as integral to enterprise risk management not back-office chores.

Unilever's 2019 disclosure of liability scenarios across six discount-rate environments set a new benchmark for corporate pension transparency, demonstrating how openness can strengthen investor trust while clarify intergenerational exposure.

Governance, here, is realism: acknowledging that promises written in 1980 still shape payrolls in 2040.

Time Preference and the African Opportunity

While aging economies wrestle with liabilities, much of Africa faces the opposite demographic equation. Nearly 60% of its population is under 25. The question isn't how to fund the past—it's how to value a rapidly expanding future.

Rwanda's **Ejo Heza** and Kenya's digital micro-pension platforms demonstrate what intergenerational fairness can look like by design. Early-stage contributions are matched, not deferred; transparency is built through mobile-first systems that track benefits in real time.

These are not side stories—they are live experiments in how societies build funded, transparent, and technologically inclusive systems before dependency ratios explode. Where the developed world discounts its future too heavily, Africa's challenge is the reverse: ensuring that the future is discounted too little.

The Ethics of Discounting

A discount rate is an ethical instrument as much as an economic one. It encodes how societies weigh today against tomorrow. Economists call it *time preference*; citizens experience it as *trust*.

When central banks hold rates near zero, they don't just stimulate economies—they rewrite generational contracts. When boards choose expected returns over risk-free rates, they don't merely smooth volatility—they redistribute the cost of time.

Favoring current retirees isn't inherently wrong if it's transparent. What corrodes legitimacy is pretending the transfer doesn't exist. Governance is less about perfection than about disclosure.

Pensions are not simply savings mechanisms—they are civilization's long-term accounting system. Every line on a balance sheet tells a story about what mattered when the math was written.

Algorithmic Actuaries and the Next Frontier

As artificial intelligence enters pension governance, discounting re-emerges in code. Algorithms now make the same ethical choices actuaries once

debated: which scenarios matter, whose future counts.

Model explainability, stress testing, and algorithmic oversight will soon test the actuarial ethics of the digital age. Boards that cannot interpret the systems risk repeating history under new terminology.

Yet technology can redeem what opacity once hid. AI-assisted scenario analysis can make intergenerational effects visible—showing how today's changes ripple through future payrolls, solvency ratios, and contribution burdens. In the right hands, algorithms can restore transparency that has been obscured.

The future of pensions will hinge less on financial engineering than on institutional humility. Systems that publish multiple valuations, tie benefits to realistic funding, and separate fiduciary judgment from political convenience will endure. Those that chase optimism through opaque math will fracture under demographic gravity.

Intergenerational fairness is not a slogan—it is the test of whether governments can measure time honestly.

Glossary

- **Discount Rate:** The assumed rate of return used to calculate the present value of future pension payments.
- **Funded Ratio:** The percentage of pension obligations covered by current assets.
- **Ultimate Forward Rate (UFR):** The long-term interest rate assumption used in pension valuations.

regulators use to value future liabilities.

- **Notional Defined Contribution (NDC):** A pay-as-you-go system that mimics defined-contribution logic by tracking each worker's virtual account.

This series explores ideas from *“Shaping the Decade: Governance, Sustainability, and AI 2026–2036,”* a guide for boards at the crossroads of governance, technology, and stakeholder capitalism. [Available Here.](#)

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